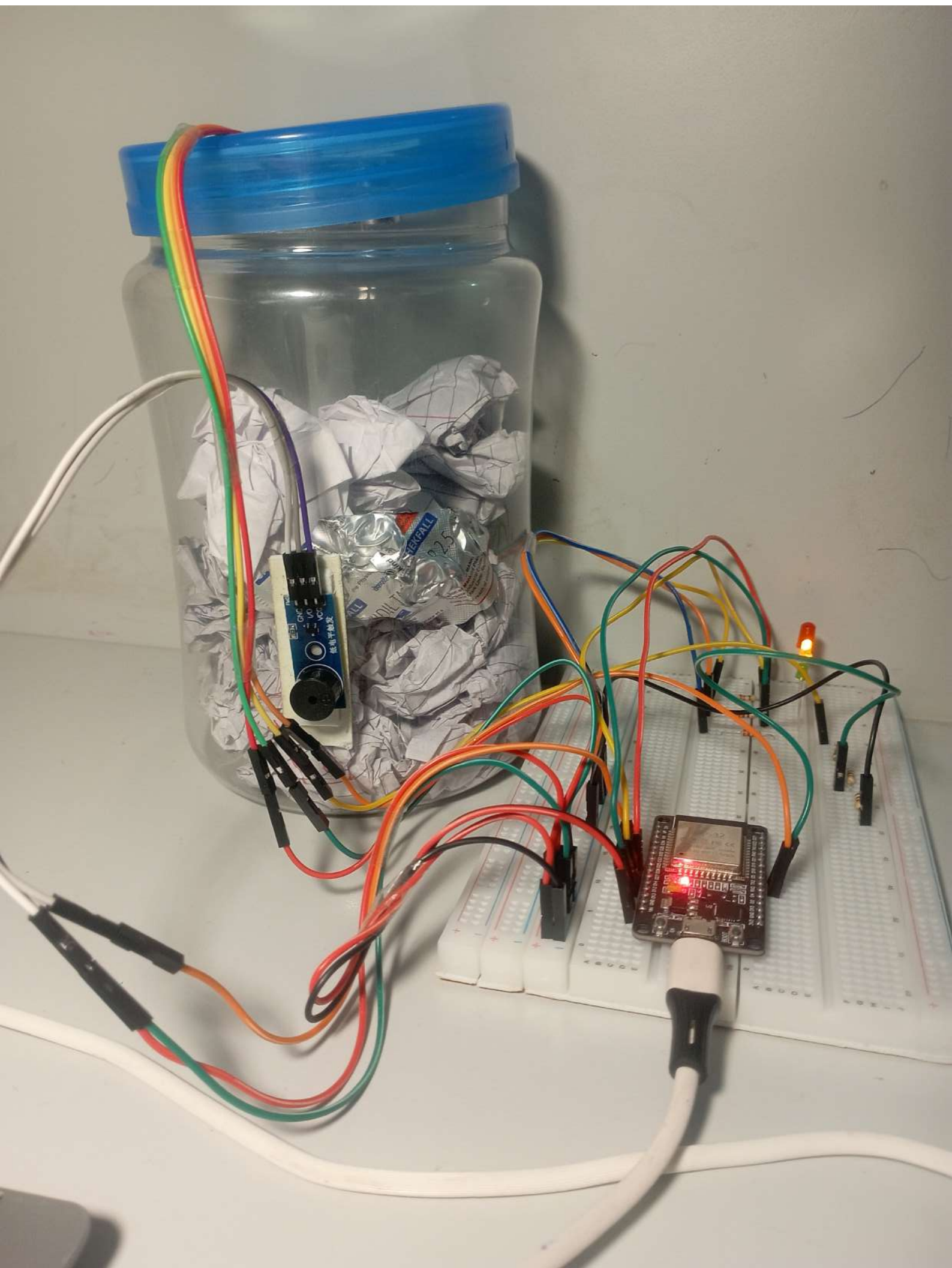
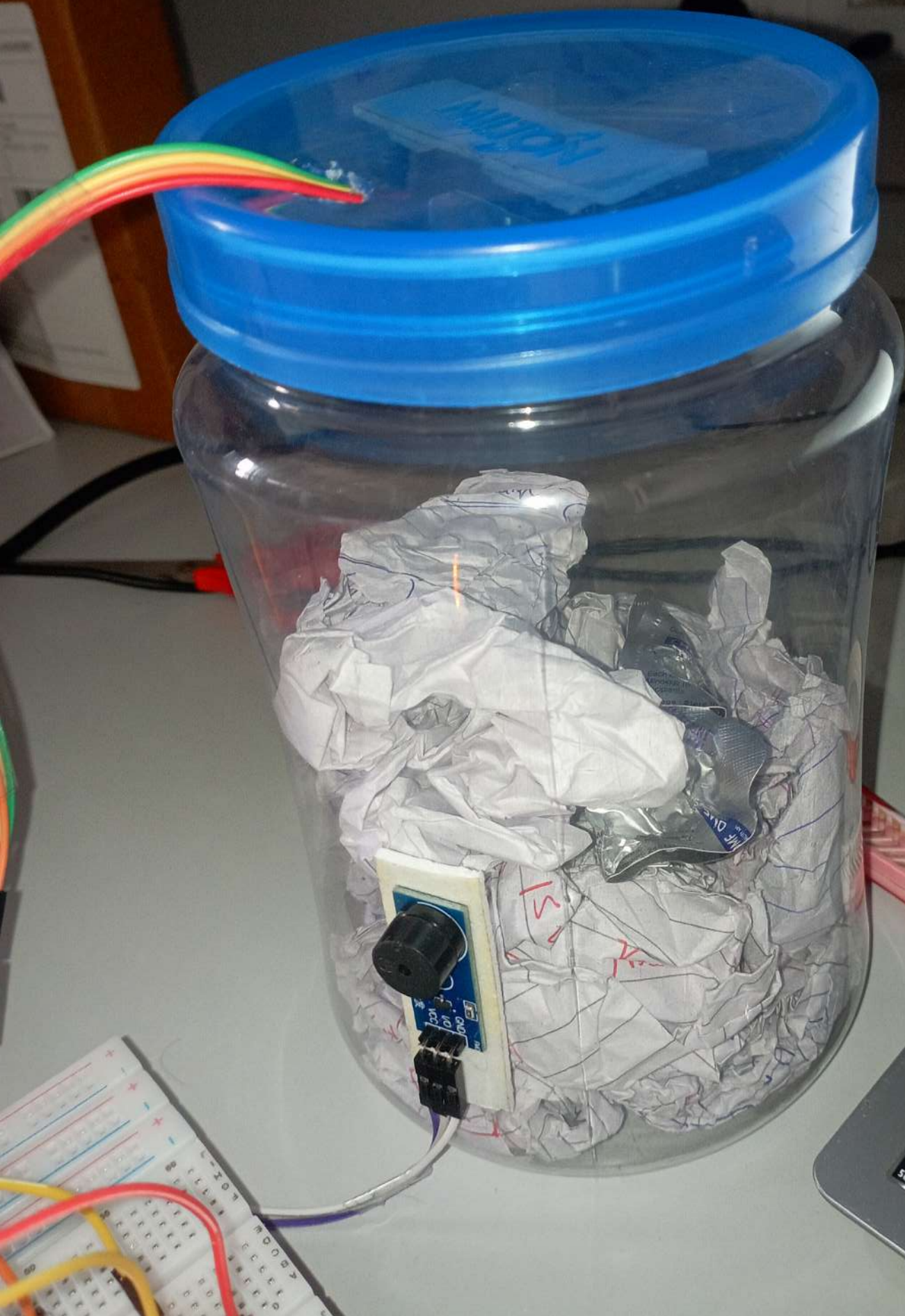
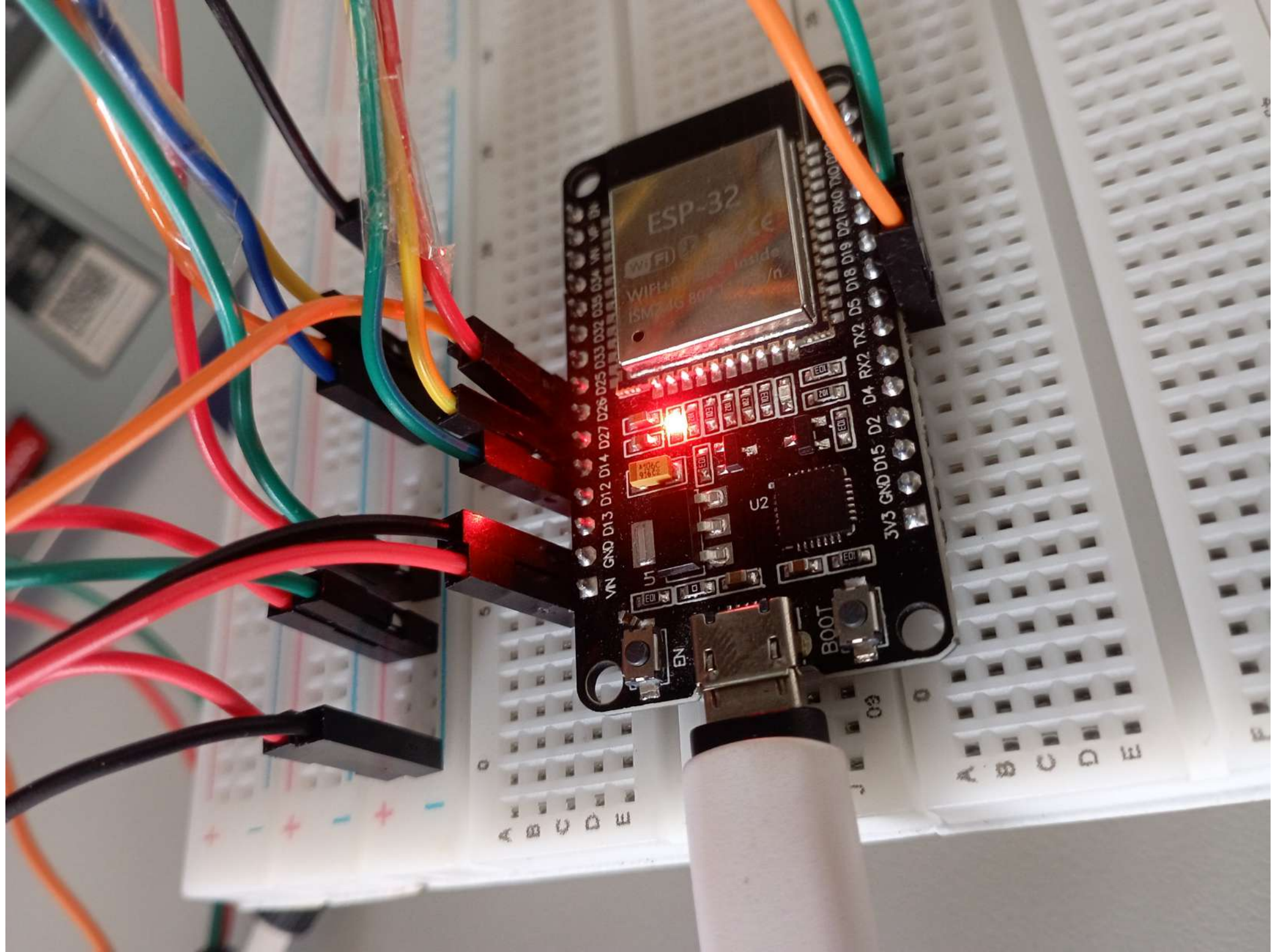
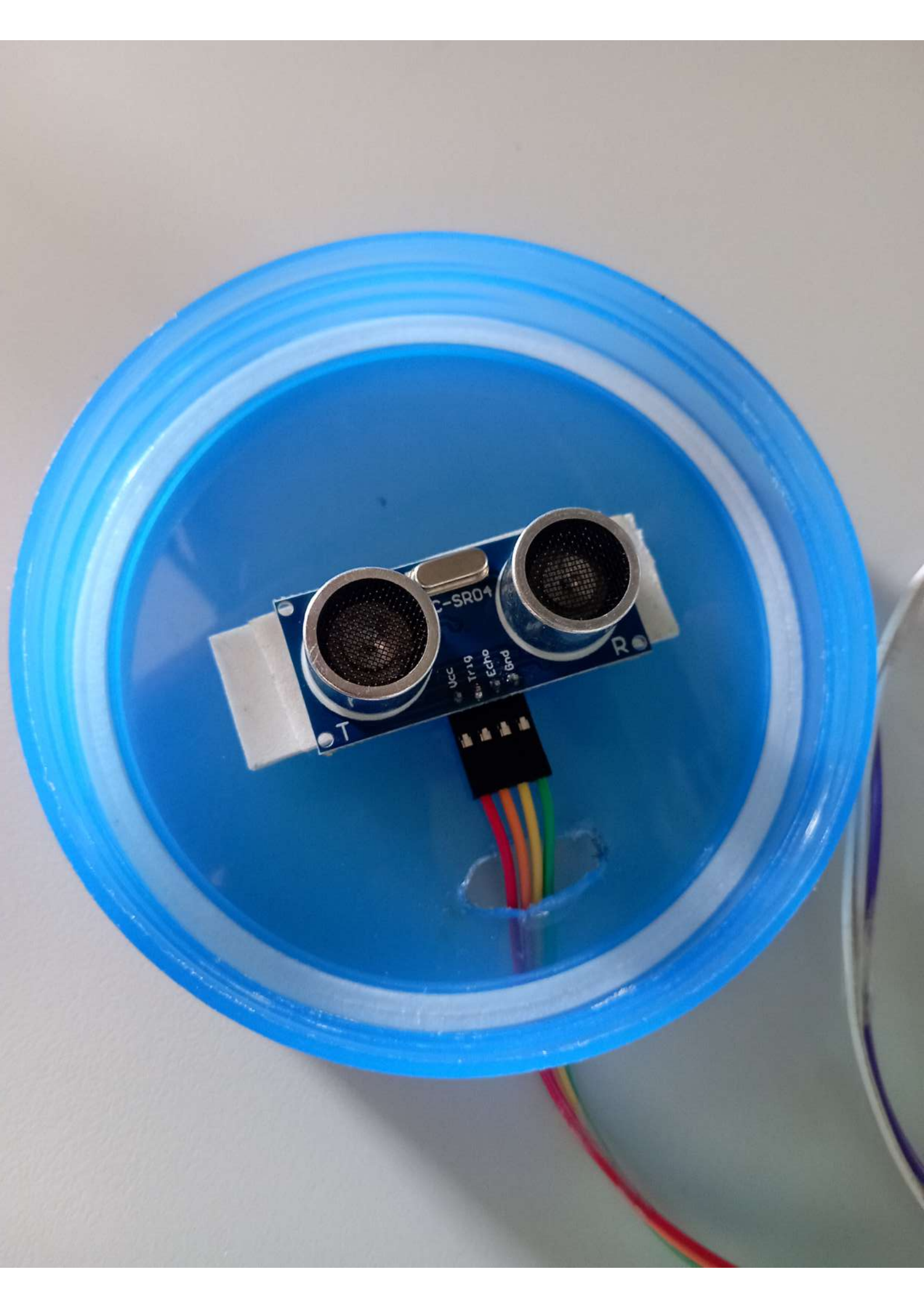


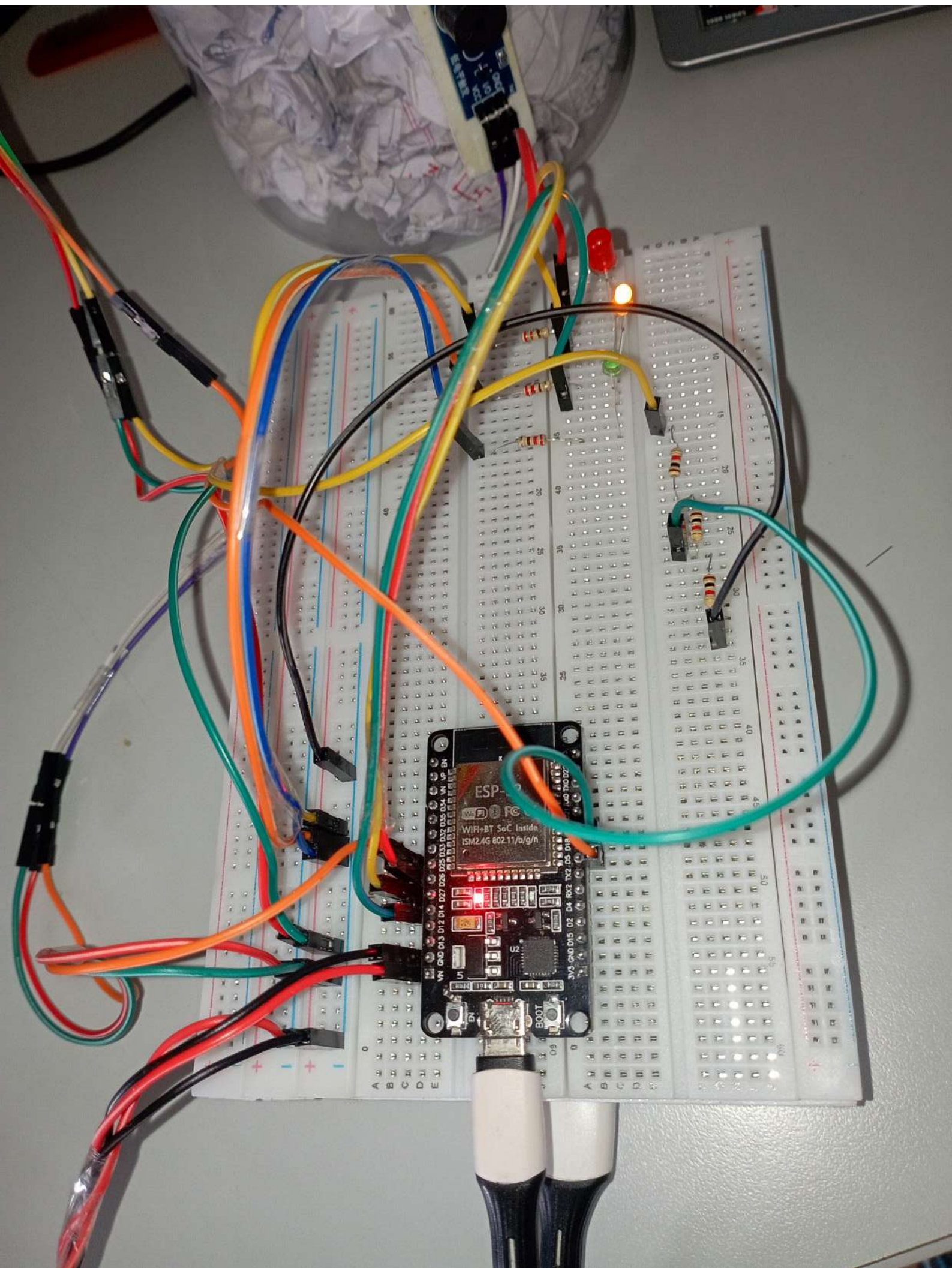


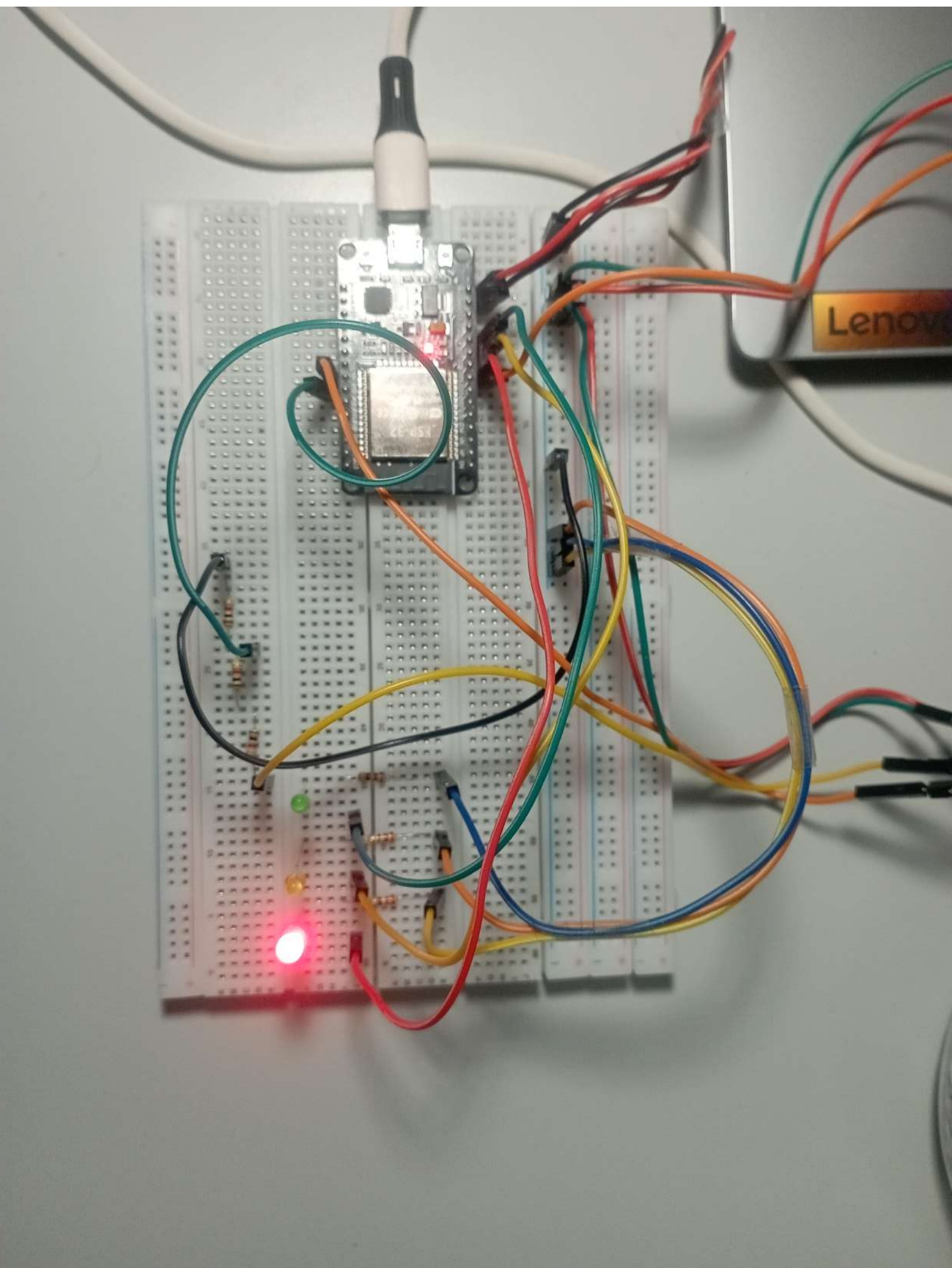


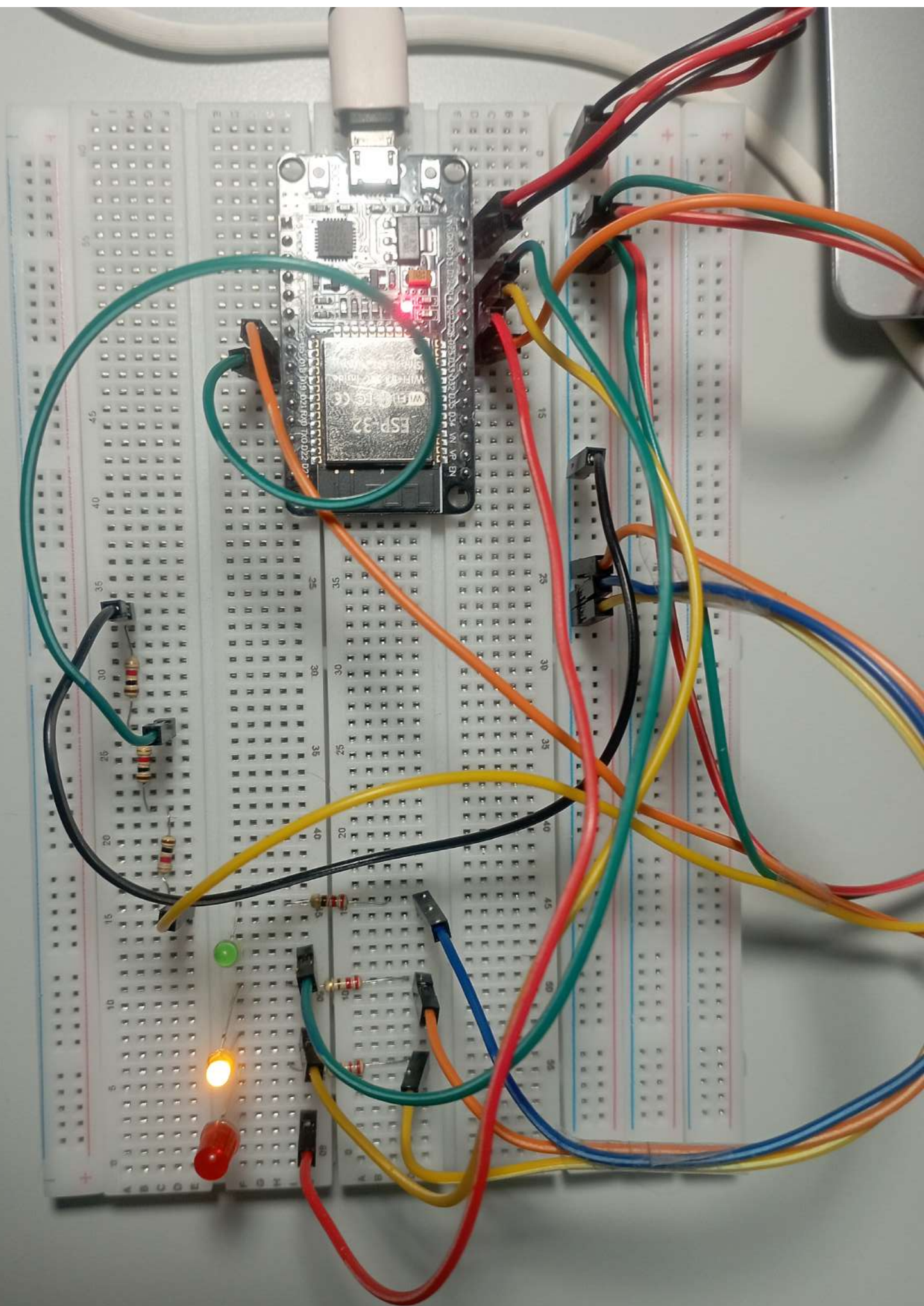


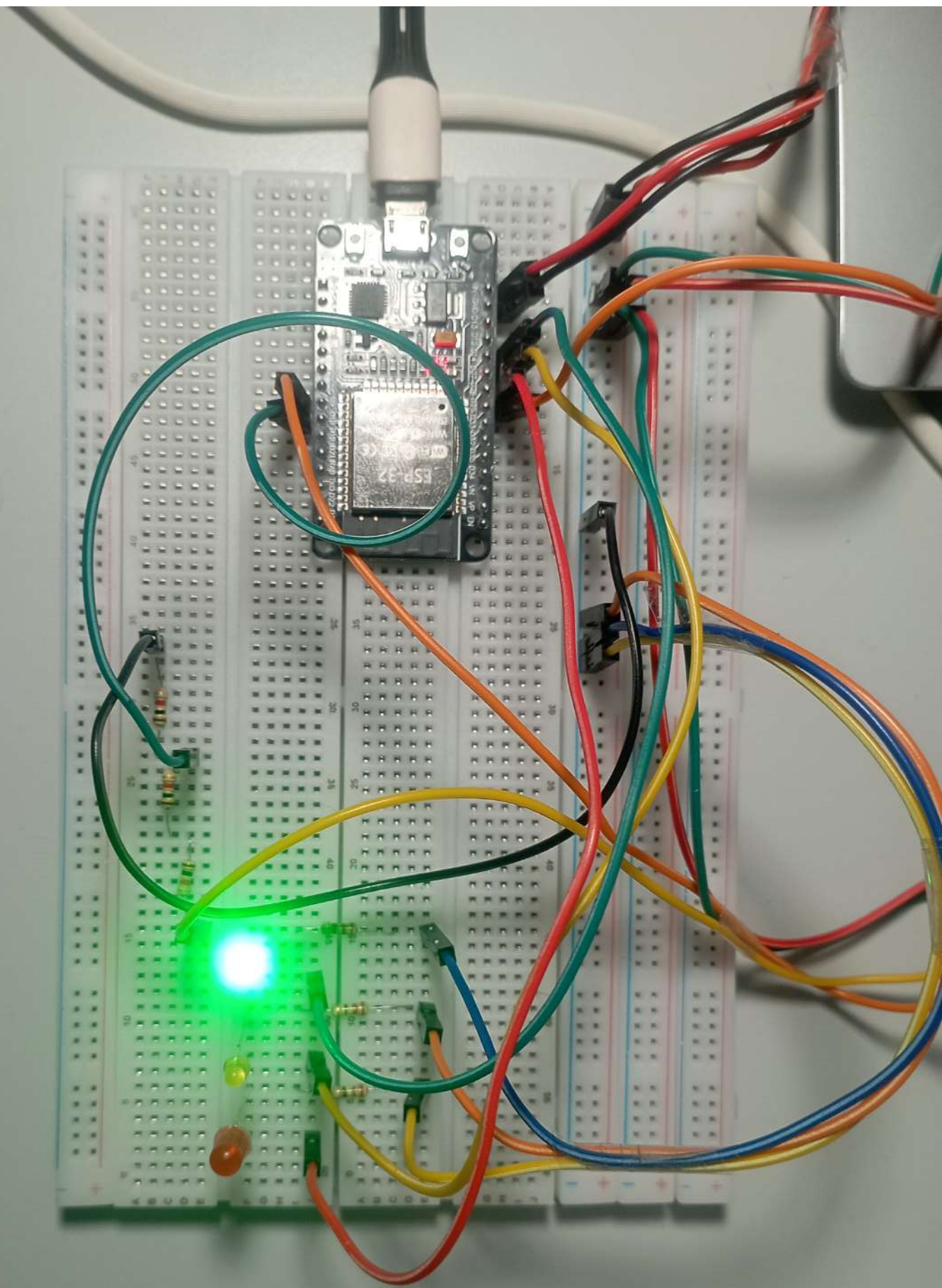


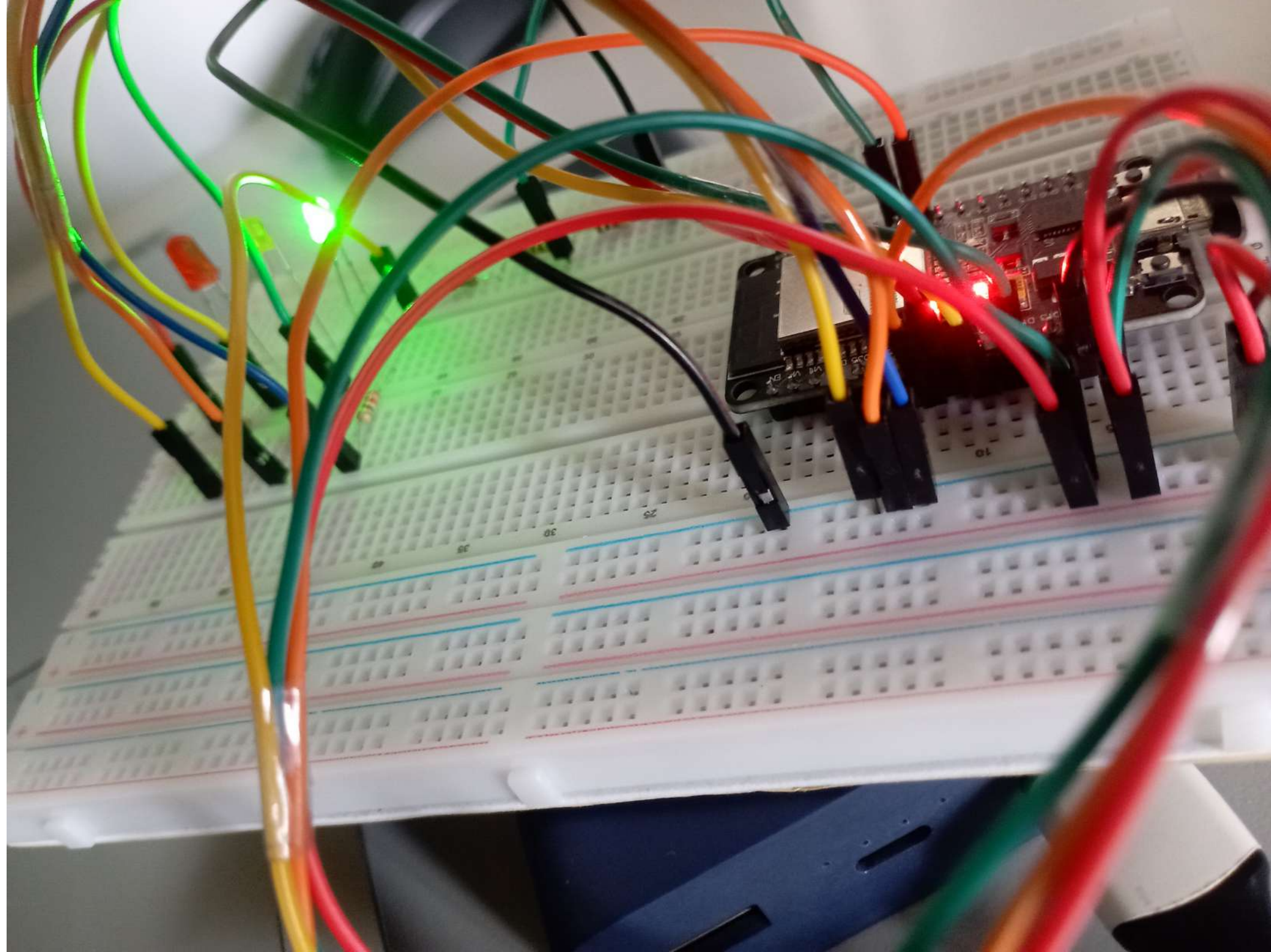


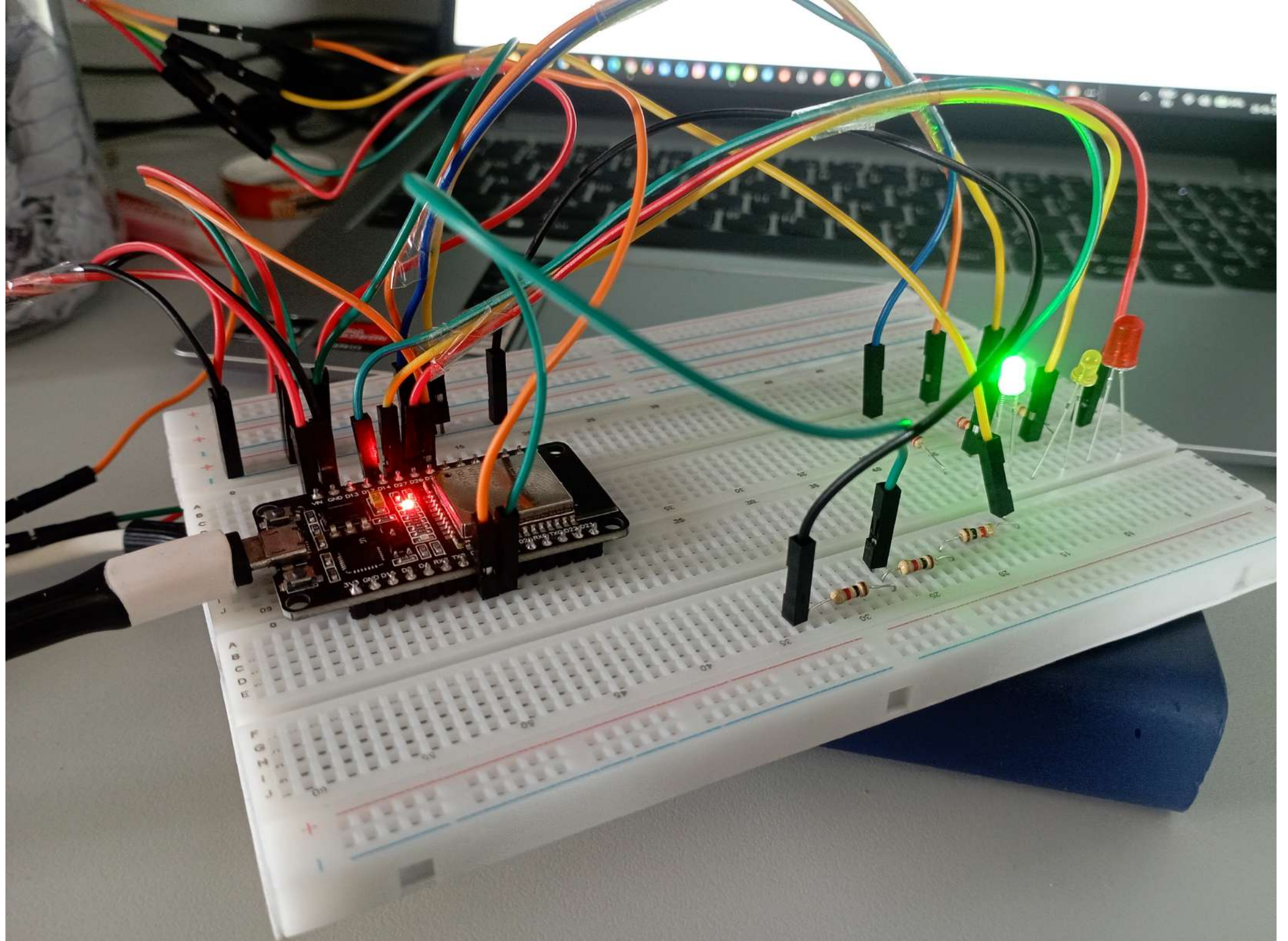


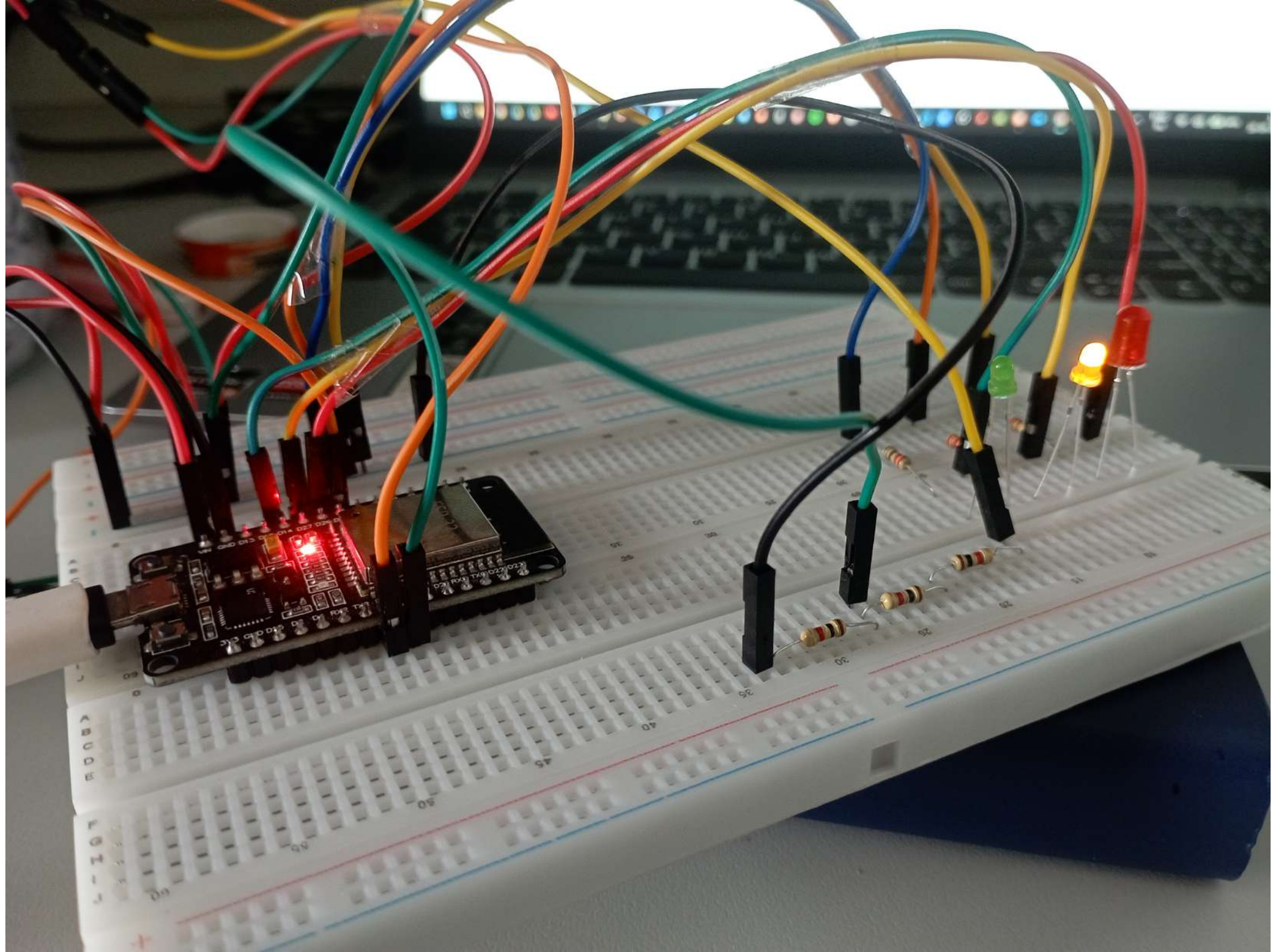












SmartBin-1

MEE2014 — GROUP 3

VIT BHOPAL UNIVERSITY



STATUS: HALF FULL ●

DISTANCE

7.7 cm

FILL LEVEL

64 %

Collection Priority:

MEDIUM

Hardware Status:

ONLINE (LOCAL AP)

Last Cleaned:

Not recorded yet

Cleaning History Log:

System booted & monitoring initialized.

 Mark as Cleaned Now

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SHIVAM SINGH & ARYAN SINGH

SmartBin-1

MEE2014 — GROUP 3

VIT BHOPAL UNIVERSITY



STATUS: OK ●

DISTANCE
10.4 cm

FILL LEVEL
39 %

Collection Priority:

LOW

Hardware Status:

ONLINE (LOCAL AP)

Last Cleaned:

Not recorded yet

Cleaning History Log:

System booted & monitoring initialized.

 Mark as Cleaned Now

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SHIVAM SINGH & ARYAN SINGH

SmartBin-1

MEE2014 — GROUP 3

VIT BHOPAL UNIVERSITY



STATUS: FULL 🚨

DISTANCE

5.8 cm

FILL LEVEL

83 %

Collection Priority:

HIGH (ACTION REQ.)

Hardware Status:

ONLINE (LOCAL AP)

Last Cleaned:

Not recorded yet

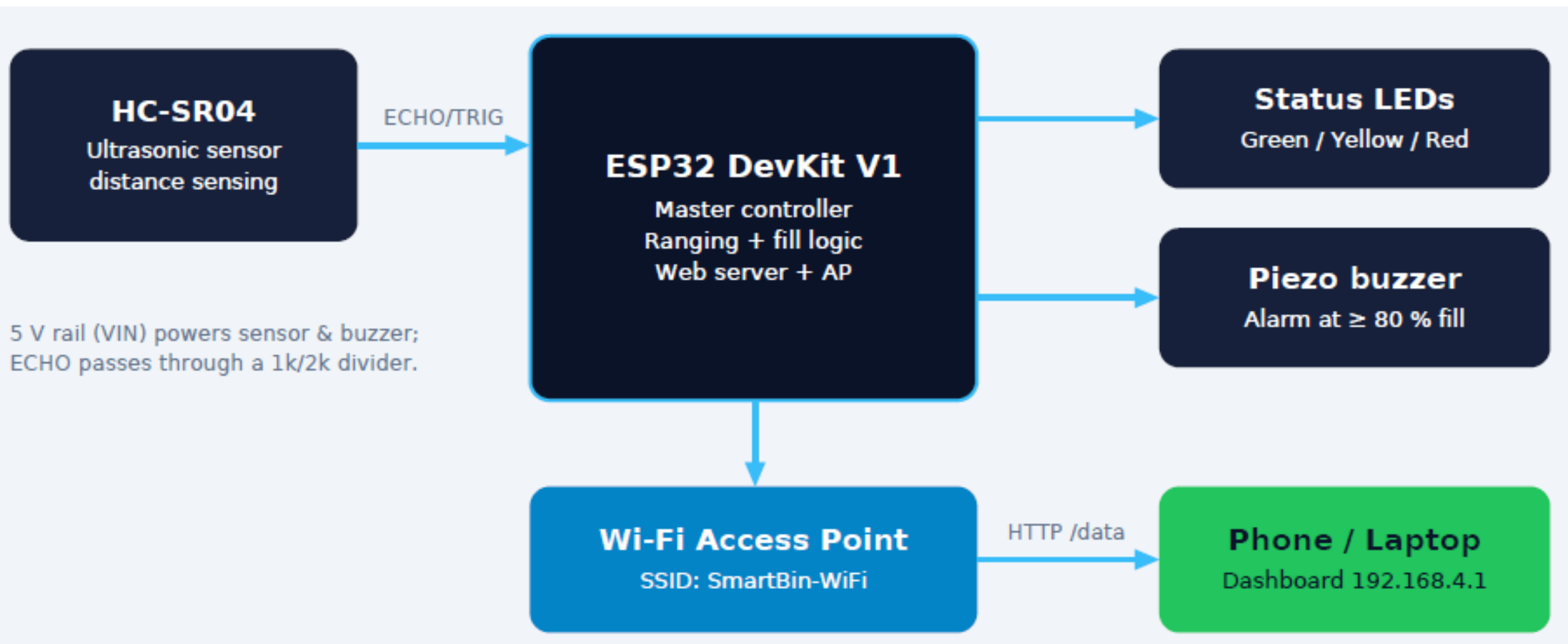
Cleaning History Log:

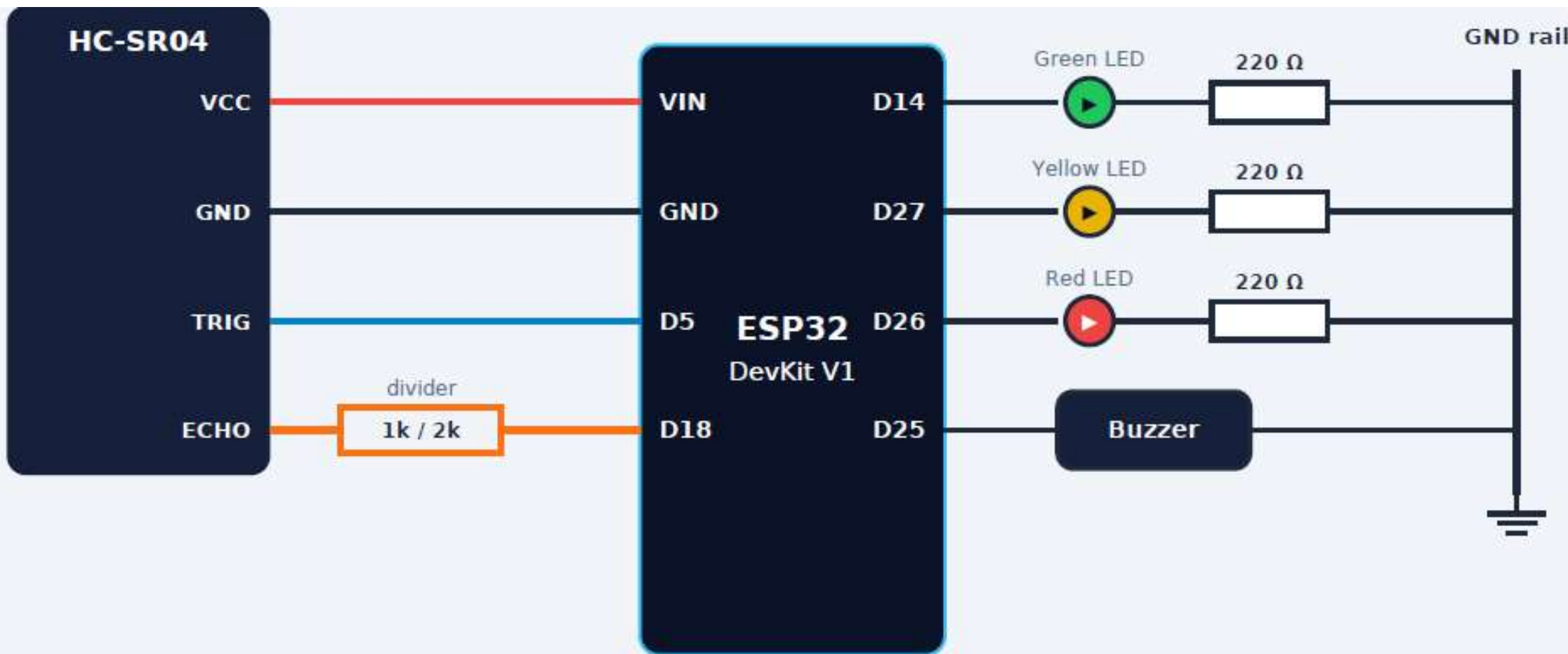
System booted & monitoring initialized.

🧹 Mark as Cleaned Now

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SHIVAM SINGH & ARYAN SINGH

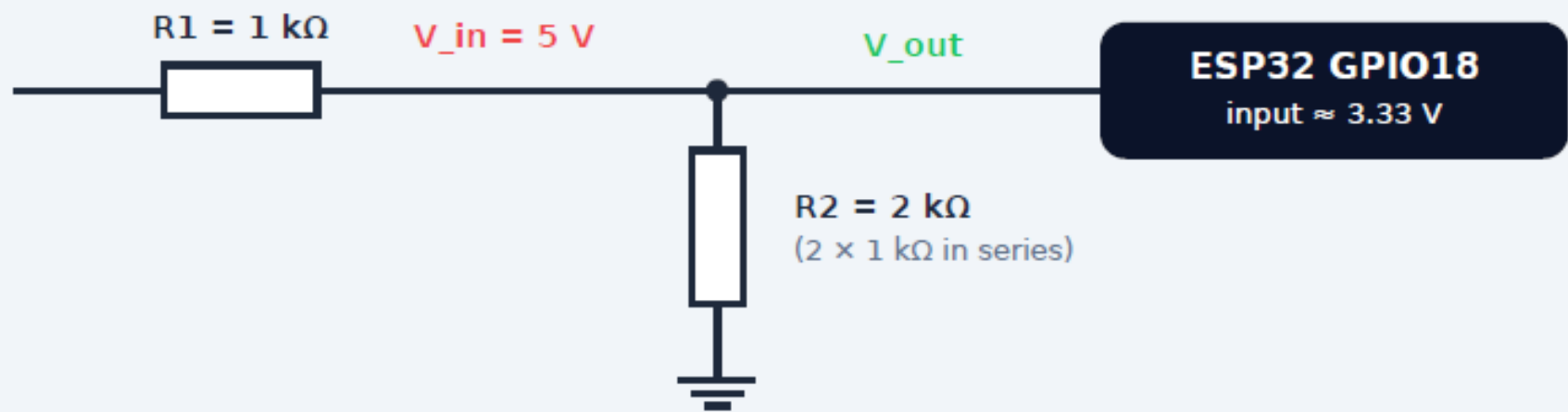




Colour key: red = 5 V • black = GND • blue = TRIG • orange = ECHO (divided to 3.3 V)

All grounds share one common rail (ESP32 GND, sensor GND, divider leg, LED resistors, buzzer –).

HC-SR04
ECHO (5 V)



$$V_{\text{out}} = V_{\text{in}} \times R2 / (R1 + R2) = 5 \text{ V} \times 2 \text{ k}\Omega / 3 \text{ k}\Omega = 3.33 \text{ V}$$